Business Problem Document

Retail Sales Performance & Profitability Analysis | Excel + SQL + Power BI

1. Background

A multi-category retailer in India sells Furniture, Office Supplies and Technology products to Consumer, Corporate and Home Office customers across five regions (North, South, East, West, Central). Between January 2023 and December 2025 sales grew every year, but the profit margin stayed flat at about 14%. Leadership suspects that growth is being bought with discounts and wants to know exactly where profit is being lost.

2. Business problem statement

Revenue is growing but profitability is not. Management cannot see which products, regions, customers or discount levels create or destroy profit, and has no single view of how performance varies by season or delivery speed. The business needs a repeatable analysis that identifies profit leaks and recommends specific actions.

3. Objectives

- Build a clean, trusted dataset from messy order exports (dates stored as text, duplicates, inconsistent text, invalid rows) using Excel.
- Measure sales, profit and margin by year, category, sub-category, region, segment and customer.
- Quantify the effect of discounting on profit and find the discount level at which orders become loss-making.
- Identify loss-making products and weak regions and explain why they underperform.
- Segment customers (RFM) and measure profit concentration.
- Deliver cleaned data, a SQL query library and an interactive Power BI dashboard, plus recommendations.

4. Key business questions

#	Question
1	What are total sales, profit, margin, orders and average order value? How are they trending year over year?
2	Which categories and sub-categories make or lose money?
3	At what discount level does profit turn negative?
4	Which regions underperform and why?
5	Is there seasonality we should plan for?
6	Who are the most valuable customers and how concentrated is profit?
7	Are delivery times acceptable in every region?

5. Data description

One flat export of order lines (one row per product per order). The dataset is **synthetic**, generated with a fixed random seed to resemble real retail order data and to include realistic data-quality problems. Period: 01-Jan-2023 to 31-Dec-2025. Currency: INR. Raw file: 9,153 rows, 18 columns.

Column	Meaning
Order ID, Order Date, Ship Date, Ship Mode	Order header information. Ship Mode: Same Day, First Class, Second Class, Standard Class
Customer ID, Customer Name, Segment	Customer identity and segment (Consumer, Corporate, Home Office)
Region, State, City	Customer geography (5 regions, 20 states)
Product ID, Category, Sub-Category, Product Name	Product hierarchy (3 categories, 13 sub-categories, 65 products)
Sales	Net sales in INR after discount = List price x Quantity x (1 - Discount)
Quantity	Units sold on the line
Discount	Discount given as a fraction (0.20 = 20%)
Profit	Sales minus product cost, in INR (negative = loss)

6. Data quality issues found in the raw file

Issue	Rows	Excel fix
Exact duplicate order lines (same Order ID + Product ID)	177	COUNTIFS duplicate flag / Remove Duplicates
Region / Segment / Customer Name casing and stray spaces	~1,100 cells	PROPER(TRIM())
Dates stored as text (dd-mm-yyyy)	all rows	DATE(VALUE(...)) / Text to Columns
Discount stored as 20 instead of 0.20	16	IF(Q2>1, Q2/100, Q2)
Missing City	109	INDEX/MATCH from the same customer's other orders
Ship Date earlier than Order Date	25	Flag and filter out
Quantity = 0	12	Flag and filter out

Result: 9,153 raw rows reduce to **8,939 clean rows** (214 removed).

7. Approach

- **Clean in Excel**: formulas on a separate Cleaning sheet, KEEP flag, filtered clean sheet; every step counted by formula.
- **Model in SQL**: customers, products, orders and order_lines tables; data-quality checks; reconciliation to the cleaned file.
- **Analyse in SQL**: 15 queries with joins, CTEs and window functions (LAG, RANK, ROW_NUMBER, NTILE), RFM segmentation.
- **Visualise in Power BI**: star schema, DAX measures, three report pages with slicers and drill-through.
- **Recommend**: turn findings into specific actions with an estimated impact.

8. Success criteria

- Every data-quality check returns zero problem rows after cleaning.
- SQL totals match the cleaned file exactly (Sales INR 53,583,538; Profit INR 7,417,634).
- Each business question has a quantified answer and an action.
- A non-technical stakeholder can use the dashboard without explanation.

9. Assumptions and limitations

- Data is synthetic: findings demonstrate the method, not real company performance.
- Profit is as reported in the source; no overhead allocation is applied.
- Orders are counted as distinct Order IDs; returns are not in the data.
- The discount-profit relationship is observational and partly mechanical (profit = price x (margin - discount)).
- The estimated uplift from a discount cap assumes volumes do not change.